

ex:- $\exists n \in \mathbb{Z}, \sim E(n)$

there exists an integer such that it's not even.

$\Rightarrow \forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y^3 = x$ is true

$$\begin{array}{r|l} 64 & 4 \\ \hline 27 & 3 \end{array}$$

$\Rightarrow \exists y \in \mathbb{R}, \forall x \in \mathbb{R}, y^3 = x$ is false

$\Rightarrow \rightarrow$ "All integers are not even" $\rightarrow \forall n \in \mathbb{Z}, \sim E(n)$
false statement.

"Not all integers are even" $\sim (\forall n \in \mathbb{Z}, E(n))$

$\Rightarrow \forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x^n \geq 0$ true